MAE 155A: Aerospace Engineering Design I

Propulsion

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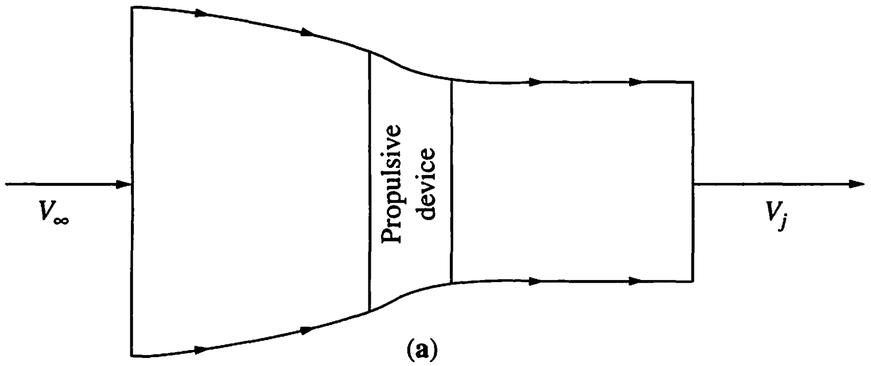
Overview

- Thrust is the net result of interactions between the surfaces of the propulsion system and the air.
- By Newton's third law, the force the air exerts on the aircraft (i.e., thrust) is equal to the force the aircraft exerts on the air, which is $T = \dot{m}(V_j - V_\infty)$, and efficiency is

$$\eta = \frac{2}{1 + V_j/V_\infty}$$

- Higher speeds lead to higher drag; therefore we need more thrust, which means higher V_j and lower η.
- Propellers are efficient but limited by wave drag (large diameter, small Δv is more efficient); jet engines apply a larger Δv to a smaller mass of air.



Anderson, John David. Aircraft performance and design. New York: McGraw-Hill, 1999.

Overview—types of propulsion systems

- Fuel-burning:
  - Chemical energy (fuel) → kinetic energy
  - Types: reciprocating engine + propeller, turboprop, turbofan, turbojet
- Electric:
  - Key components: battery, generator, inverter, motor, propeller
  - Pure electric:
    - Chemical energy (battery) → electrical energy → kinetic energy
  - Turbo-electric:
    - Chemical energy (fuel) → kinetic energy → electrical energy → kinetic energy
  - Hybrid electric:
    - Chemical energy (battery, fuel) → … → kinetic energy

Basic thrust analysis

Control volume's surfaces include:

- Overall inlet + exit faces (where arrows for V_∞ and V_j are labeled)
- Inlet + exit faces of the propulsive device
- Outer sheet of streamlines

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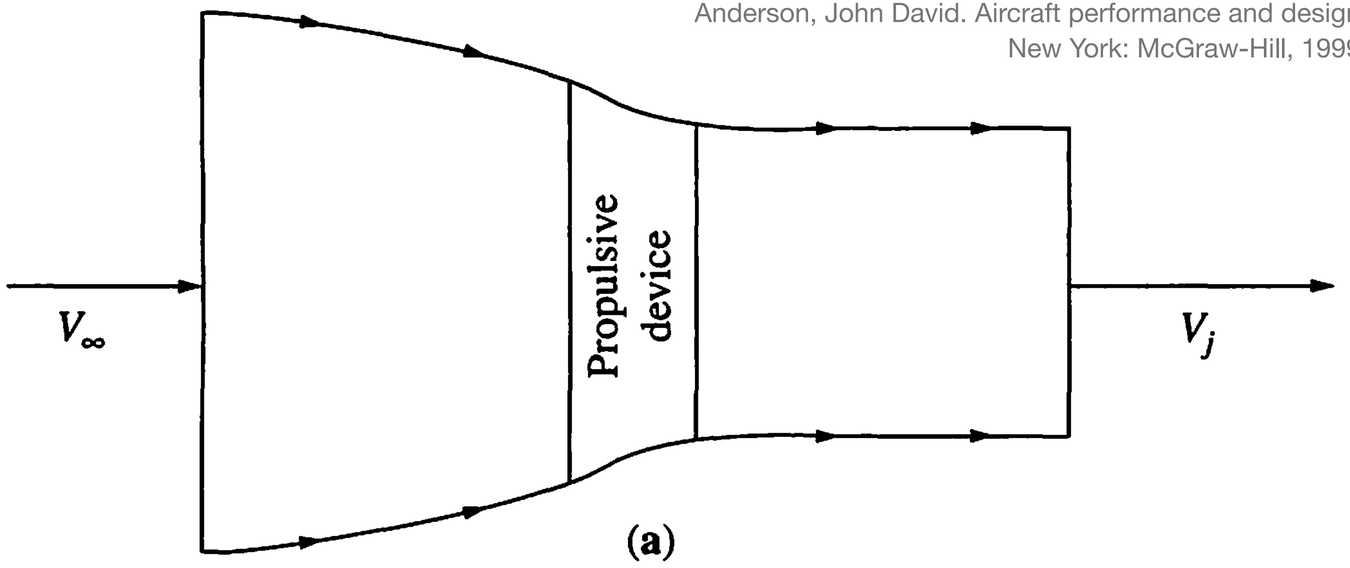


Conservation of momentum (x)

$$\frac{d}{dt}\int_{CV} \rho u \, dV = -\int_{CS} \rho u (\mathbf{V}\cdot\mathbf{n}) \, dA + \sum_i F_x^{(i)}$$

(Accumulation of x-momentum is amount flowing in minus out plus sources/sinks (forces))

$$0 = -\rho V_\infty(-V_\infty)S_{inlet} - \rho V_j(V_j)S_{exit} + T \quad \Longrightarrow \quad 0 = \dot{m}V_\infty - \dot{m}V_j + T.$$

$$\Longrightarrow T = \dot{m}\Delta V = \dot{m}(V_j - V_\infty) \text{ where } \quad \dot{m} = \rho S V$$

Basic thrust, power, and efficiency analysis

We are interested in **propulsive efficiency**—the efficiency of the air in propelling the aircraft forward. Efficiency is always a ratio of output to input, where the output is always lower due to losses.

(1) The output power (P_{out}) is the useful propulsive power. It is the rate at which work is being done by the air in pushing the aircraft forward. Sometimes, we call this thrust power.

$$P_{out} = TV_\infty \quad \Longrightarrow \quad P_{out} = \dot{m}(V_j - V_\infty)V_\infty \quad \text{(since } T = \dot{m}(V_j - V_\infty)\text{)}$$

(2) The input power (P_{in}) is the rate at which the propulsive device does work on the air. We apply conservation of energy to the control volume, which tells us that the input power is the difference between the rate of kinetic energy flowing out at the outlet and that at the inlet (the rate at which the air's kinetic energy increases):

$$P_{in} = \frac{1}{2}\dot{m}V_j^2 - \frac{1}{2}\dot{m}V_\infty^2 \quad \Longrightarrow \quad P_{in} = \frac{1}{2}\dot{m}(V_j + V_\infty)(V_j - V_\infty)$$

$$\eta = \frac{P_{out}}{P_{in}} = \frac{2}{1 + V_j/V_\infty}$$

Note: we can also derive this via $P_{in} = P_{out} + P_{loss}$ with $P_{loss} = \frac{1}{2}\dot{m}(V_j - V_\infty)^2$

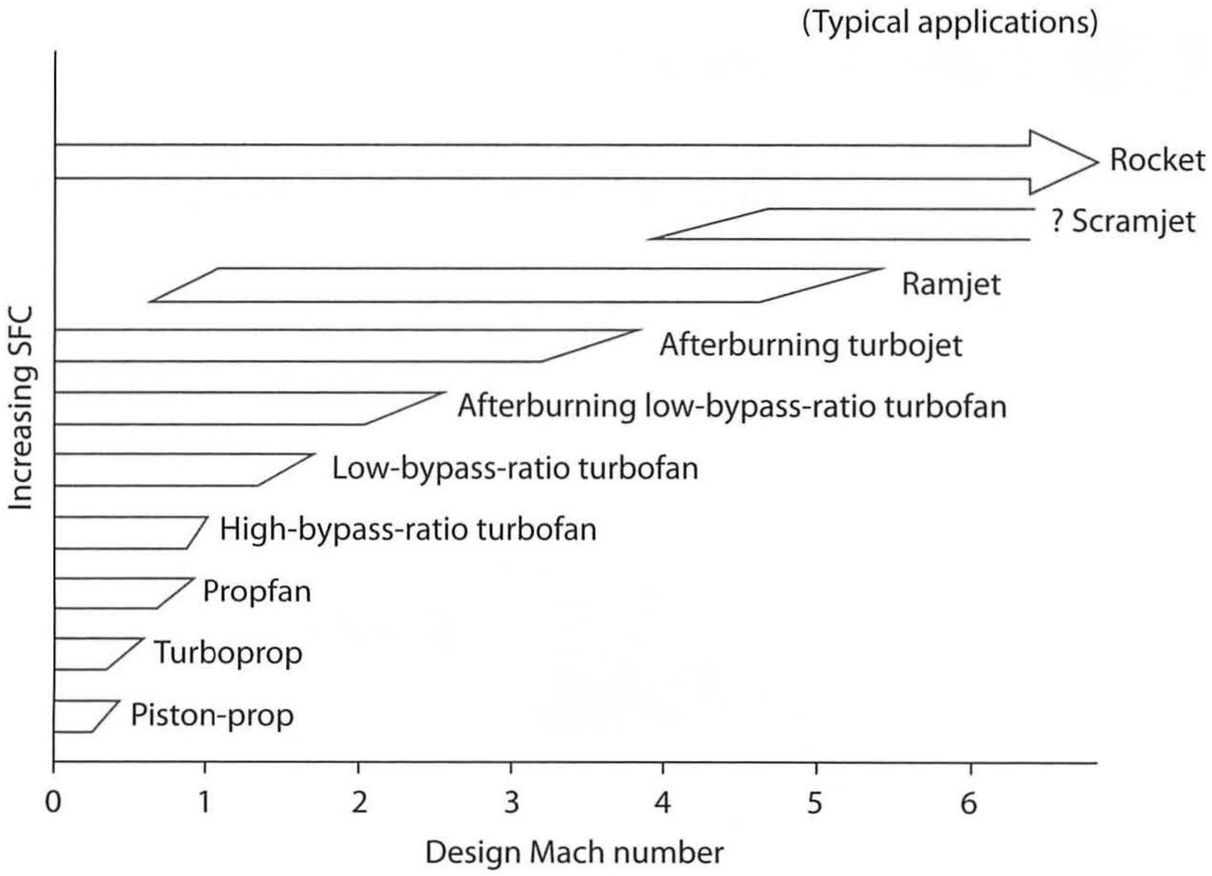
Basic thrust, power, and efficiency analysis

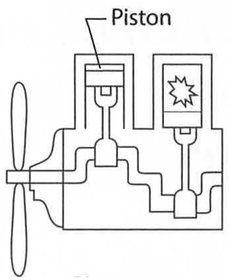
$$\eta = \frac{P_{out}}{P_{in}} = \frac{2}{V_j/V_\infty + 1}$$

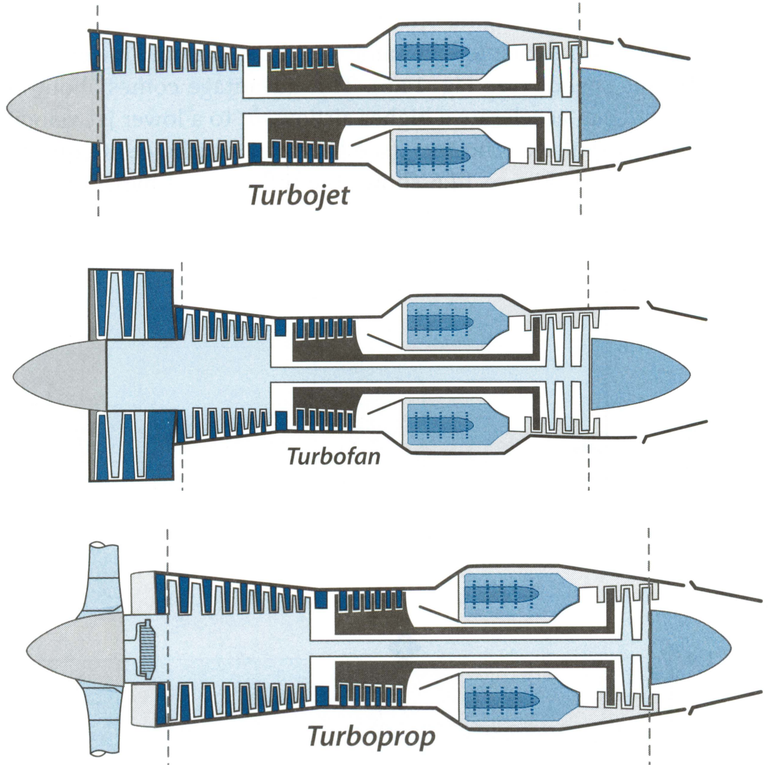
$T = \dot{m}(V_j - V_\infty)$ where $\dot{m} = \rho S V$

- We must satisfy $V_j > V_\infty$ in order to generate thrust
- As $V_j \to \infty$, $\eta \to 0$
- As $V_j \to V_\infty$, $\eta \to 1$
- Therefore, we want V_j to be as small (close to V_∞) as possible for maximum propulsive efficiency
- This means that, for the same amount of thrust} T, S must be as large as possible

Fuel-burning propulsion







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Turbine engines

- Four steps: compress air; mix it with fuel; burn the mixture; extract energy from the hot, high-pressure gas
- Pros: can operate at higher speeds
- Cons: not as efficient, especially at low speeds, due to small mass flow and high exhaust velocity
- Thrust is approximately constant with speed and increases linearly with density
- Turbofan and turbojet: we look at thrust and thrust-specific fuel consumption (TSFC) versus speed/Mach and density (formulas shown in a later slide)
- Turboprop: we look at power and brake-specific fuel consumption (BSFC) versus speed/Mach and density (formulas shown in a later slide)

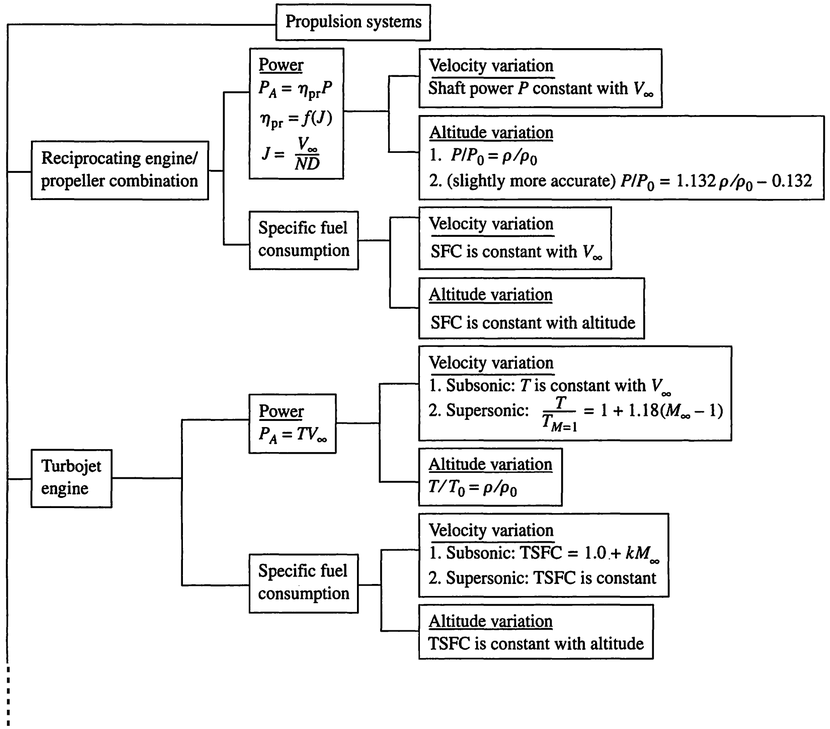
$$TSFC = c_t = \frac{\dot{m}_{fuel}}{T}, \quad BSFC = c_a = \frac{\dot{m}_{fuel}}{P}$$

Turbine engines

- Turbojet engines:
  - Compressor: compresses air to many times atmospheric pressure.
  - Burner: injects fuel, mixes it with compressed air, and burns the mixture to increase temperature.
  - Turbine: extracts energy from the hot gas to drive the compressor before expanding in exhaust nozzle.
- Turbofan engines:
  - Idea: turbine drives a ducted fan; part of the air flow bypasses the core (compressor, combustor, turbine).
  - Efficient and quiet: the duct suppresses noise, and the thrust from the bypassed air enables a reduced exit velocity because of reduced strength of noise-generating shear layers.
- Turboprop engines:
  - Idea: optimize the turbine to drive a propeller that generates the bulk of the thrust, rather than the jet.
  - Prop-fan or unducted fan: advanced, high-speed propellers that are efficient but loud.
  - Open rotor: modern prop-fans that are large, counter-rotating; even more efficient but louder.

Fuel-burning propulsion scaling laws

- Estimate sea-level static thrust or power from W_0 and T/W or P/W.
- Given these values, you can estimate thrust or power at a given altitude and speed or Mach number using the equations below. The same goes for specific fuel consumption.



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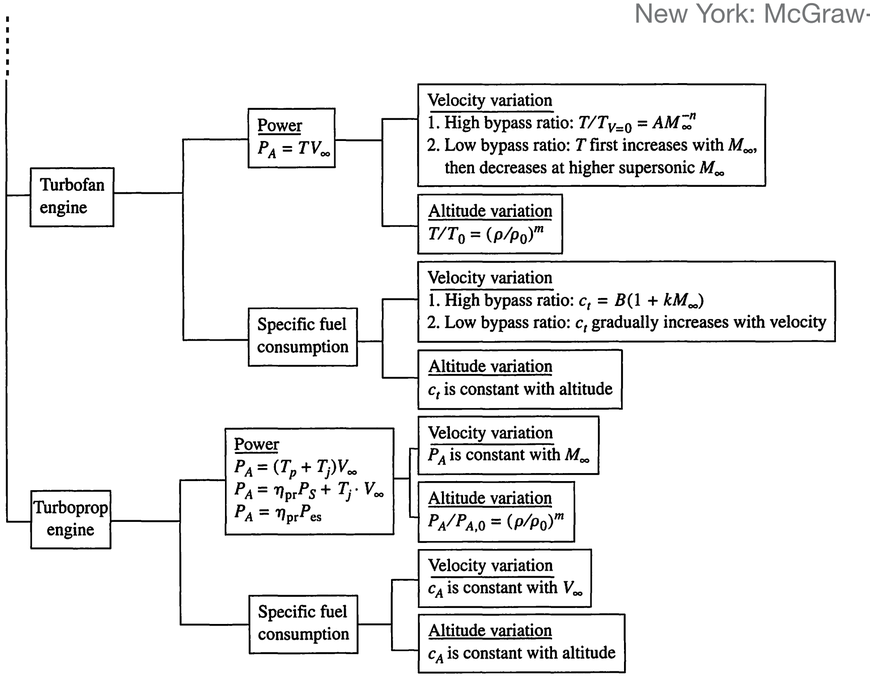


Computing atmospheric properties

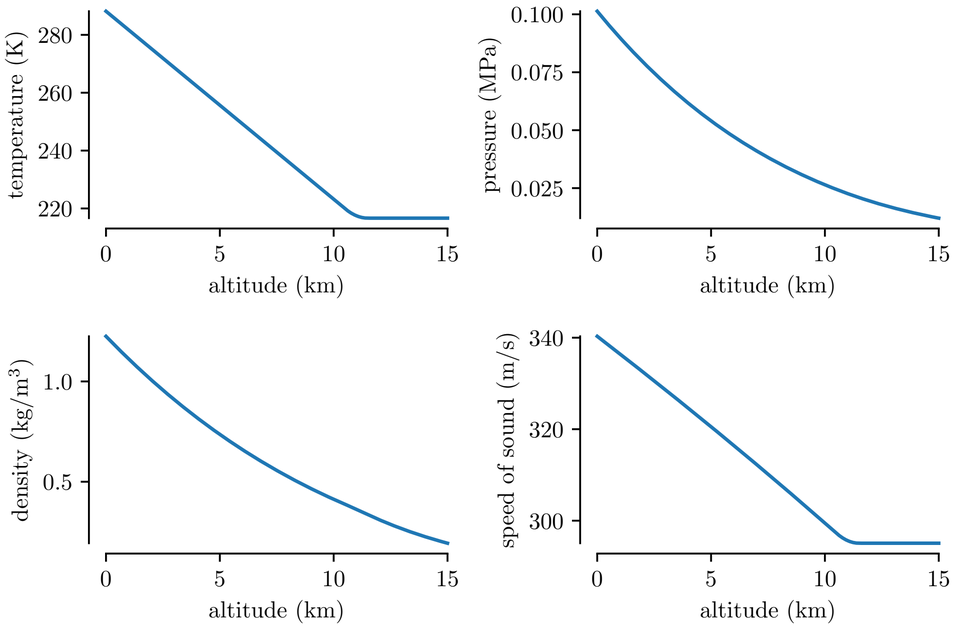
You can estimate temperature, pressure, density, and speed of sound as a function of altitude using simple analytical equations.

Temperature is approximated to be piecewise linear (break at ~11 km).

Pressure can be computed from the hydrostatic equation ($\frac{\partial P}{\partial h} = -g\rho$).

Density can be computed using the ideal gas law ($P = \rho RT$).

The speed of sound can be computed from $a = \sqrt{\gamma RT}$.



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